

Assignment – RAG & Vector Store Fundamentals

Choose the single best answer for Q1–Q4 | Q5: Submit source code for one project

Q1 · Chunking

Q1. What is the primary reason for chunking a document before storing it in a vector store?

- (A) To reduce the file size of the original document
- (B) To split the document into smaller pieces that fit within the LLM's context window
- (C) To convert the document into a different file format
- (D) To remove duplicate content from the document

Q2 · Vector Store

Q2. What does an embedding model produce when given a piece of text?

- (A) A compressed ZIP version of the text
- (B) A keyword index of the most important words
- (C) A numerical vector that represents the meaning of the text
- (D) A summary of the text in fewer words

Q3 · RAG – Retrieval

Q3. In a RAG pipeline, what happens after the user's query is converted into an embedding?

- (A) The query is sent directly to the LLM for an answer
- (B) The query embedding is compared against stored vectors to find the most relevant chunks
- (C) The original documents are re-read from scratch each time
- (D) The query is stored in the vector store permanently

Q4 · RAG – Generation

Q4. What is the correct order of steps in a basic RAG pipeline?

- (A) Generate → Retrieve → Embed → Store
- (B) Store → Retrieve → Embed → Generate
- (C) Embed → Store → Retrieve → Generate
- (D) Retrieve → Embed → Store → Generate

Q5 · Mini Project

Q5. As part of the practical evaluation for this module, each student is required to independently develop one AI-powered application from the options listed below. Students must submit their source code.

1. Resume Analyzer
2. Chatbot
3. Email Generator
4. Content Generator